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RTX 3080 performance and project readiness

Psychology VR | Detailed technical report | 17 October 2026 | Source release v0.5.4

Overall assessment: the test kit is usable and the individual components work in the tested cases. The requested complete, reliable, <=14-second spoken conversation has **not** been demonstrated. The 10 GB GPU memory limit and dialogue offload are the central constraints.

The strongest result is fast optional QAT Q4 dialogue when its compute blocks fit on the GPU. The largest obstacle is keeping that dialogue model, the approved BF16 cloned-speech model and the game resident together. A successful voice-only run does not establish that the combined application will have the same speed.

Finding	Measured result	Assessment
Software regression suite	204 tests passed	Functional coverage passed; no latency certification
Optional QAT dialogue only	3.681 s warm median	Promising component result
Selected BF16 voice clips	4.233 s warm median	Works alone; memory is very tight
CPU STT, beam 1	3.042 s median	Works on the selected recording
Sampled quick-test GPU peak	9,922 / 10,240 MiB	96.9% of device memory in use
Optional NF4 combined run	31.955 s first prepared reply; next turn failed	Failed the performance/reliability screen
Actual audible <=14 s result	Not measured	Acceptance remains open

What this means for the project

There is meaningful progress in model integration, reproducible measurements and diagnostics. The current evidence supports continued engineering and controlled demonstrations. It does not support claiming a consistently fast, fully validated training application on a 3080.

This report explains the completed 17 October measurements. It adds no new 4090 benchmark. Results labelled measured came from the local 3080 10 GB; 12 GB and 4090 outcomes remain unverified. [E1-E5]

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Test environment and definitions

Keep the tested machine distinct from the intended deployment.

Item	Actual test environment	Intended target
CPU	Intel Core i9-14900K, 24 cores / 32 logical processors	14th-generation Core i9; exact target SKU unconfirmed
System RAM	32 GB installed; 31.77 GiB visible	64 GB requested
GPU	RTX 3080, 10,240 MiB	RTX 4090, 24 GB
NVIDIA driver	617.14 in the recorded test	Target driver not captured
Other capacity	3080 12 GB profiles are provided	No local 12 GB measurements
Game and headset	Excluded from component benchmarks	Must be tested in desktop, then VR if required

Recorded software

Python 3.12.13; PyTorch 2.8.0+cu128; Transformers 5.17.0; faster-whisper 1.2.1; CTranslate2 4.8.2; bitsandbytes 0.50.2. Dialogue uses bundled CUDA llama.cpp b10909. Lip alignment uses Rhubarb 1.14.0. These versions describe the tested installation, not a recommendation to upgrade everything.

Timing definitions

Term	What the report measures
Model load / startup	Loading weights and bringing a worker online; kept separate from reply timing. BF16 voice load was 14.009 s in the new run.
Dialogue only	Real appraisal plus response generation, context handling and validation. No STT, TTS or audible playback.
Voice only	Preparation of a complete test WAV from fixed text and assigned reference. It includes the enabled delivery/validation behavior.
First prepared reply	Time until the benchmark sees a ready audio segment. With an input WAV it includes transcription. It is not first audible sound.
End-to-end audible	End of user speaking to sound from the actual game/audio output. This is the requested target and was not measured here.

Quick and tuning dialogue runs contain four requests each. Voice testing contains four characters and two passes. STT compares two repetitions per beam on one recording. These are screening samples; their p90 figures are descriptive interpolation, not reliable tail-latency estimates. [E1, E3]

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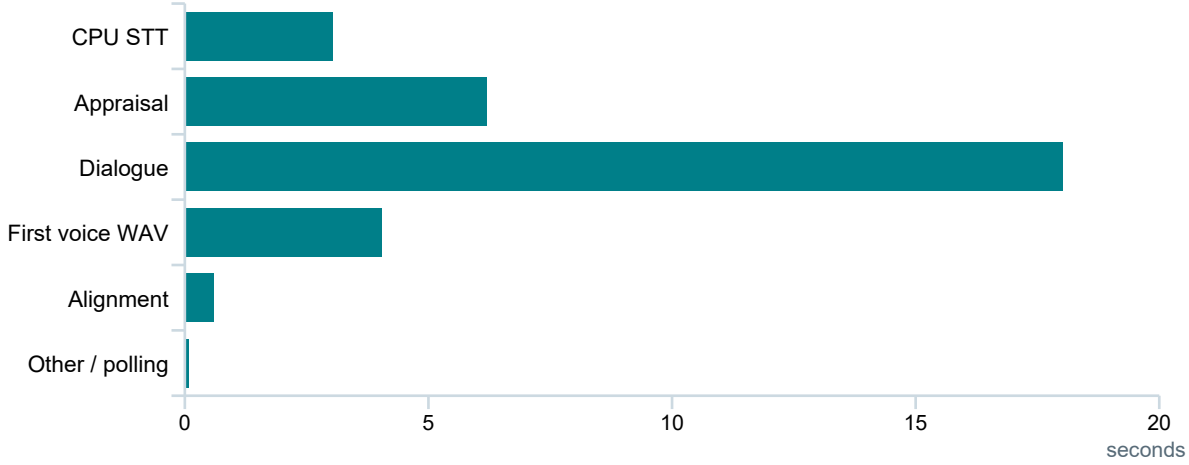
Where a reply spends its time

The architecture contains several serial steps before the first voice segment.

User stops speaking -> endpoint detection -> CPU STT -> appraisal -> dialogue generation -> reply/language checks -> first voice WAV -> mouth alignment -> download/playback.

Appraisal and dialogue are separate model calls in the current bridge. Voice can then be delivered as completed sentence segments. The first sentence still waits for the generated response and its own synthesis/alignment work. This is not continuous LLM-token-to-PCM streaming.

Reconstruction of the measured first NF4 combined reply



Phase	Seconds	Share of first-ready wait
STT	3.036	9.5%
Appraisal + dialogue	24.223	75.8%
First speech synthesis	4.033	12.6%
First alignment + residual overhead	0.662	2.1%
Total first prepared reply	31.955	100%

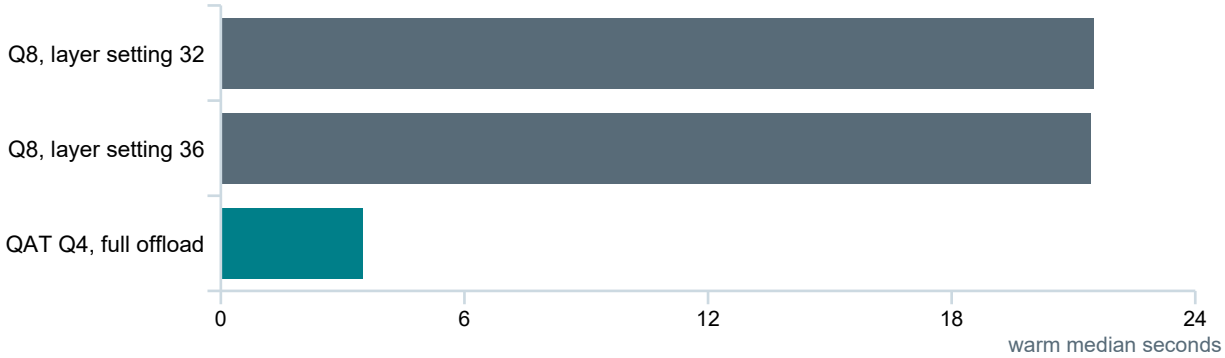
This is an approximate reconstruction from recorded stage timings; small residuals include polling and surrounding work. Appraisal plus dialogue account for about 75.8% of this wait. Saving a fraction of a second in STT or alignment cannot by itself solve that run. The game, microphone endpointing and transport are still excluded. [E1, E3]

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Dialogue performance and offload

The largest measured improvement came from a different model/offload configuration.

Earlier same-campaign comparison



Warm medians were 21.495 s for Q8 at setting 32, 21.413 s at setting 36, and 3.486 s for fully offloaded QAT Q4. The 32-to-full-QAT comparison is an observed 6.17x configuration speed ratio. It changes precision, offload and potentially answer length; it does not isolate a universal 6.17x quantization benefit. Q8 setting 36 also had a 38.286 s first request. [E2]

New quick-test dialogue details

Turn	Total s	Appraisal s	Dialogue s	Spoken words
1	3.873	0.983	2.826	12
2	3.589	0.965	2.605	27
3	3.681	0.930	2.730	39
4	4.019	1.104	2.889	37

All four requests completed with one appraisal call and one dialogue call. Overall median was 3.777 s, first request 3.873 s, warm median 3.681 s and maximum 4.019 s. The ordinary target is 18 words, but open/detail questions can legitimately use the fuller relationship budget. The 27-39-word answers above show why 18 must not be described as a universal hard cap.

What a GPU-layer number actually means

In bundled b10909, the output layer is included in the offload count. Gemma has 48 transformer blocks: setting 16 means 15 blocks plus output on GPU, leaving 33 blocks on CPU; 32 means 31 plus output, leaving 17; 36 means 35 plus output, leaving 13. Setting 49 offloads all eligible blocks/output; 99 requests all. Some host/input work remains. This explains how a busy GPU and moderate total CPU utilization can coexist with slow token generation. [R1]

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Cloned-voice performance

Four assigned speakers; BF16, sentence mode, optional decoder adapter, unused-cache release.

Character	First pass s	Warm pass s	Audio length s	Warm RTF
Alex	8.288	3.963	3.72	1.065
Ivan	3.387	3.469	3.36	1.032
Stefan	6.913	4.503	4.48	1.005
Nikola	7.418	4.583	4.56	1.005

Across eight clips the median was 4.543 s, p90 7.679 s and maximum 8.288 s. Across only the four warm clips the median was 4.233 s and maximum 4.583 s. BF16 model loading took 14.009 s, outside these synthesis timings. [E1]

Real-time factor explains the later gaps

RTF = synthesis time / generated audio duration. RTF below 1 means faster-than-playback generation for that sample; above 1 means slower. These warm clips were around 1.005-1.065. That leaves little production-speed margin once alignment, longer text, contention or a retry is added. A system can start an answer acceptably yet pause before its next part.

The voice-only test sends two-sentence text and waits for the full WAV; the app can split delivery into separate segments. Sentence segmentation can improve first-segment readiness, but adds repeated prompt/reference processing and may change prosody. Its advantage must be judged from actual first sound and later gaps, not just total WAV generation time.

What has and has not been checked

- Each character used its own routed reference and seeded recipe. The first and warm WAV hashes matched for all four test texts, showing sample repeatability in this run.

- Stefan requested sadness but resolved to his own approved neutral fallback in the recorded voice log. A neutral fallback preserves speaker identity, but is not evidence of a validated sad performance.

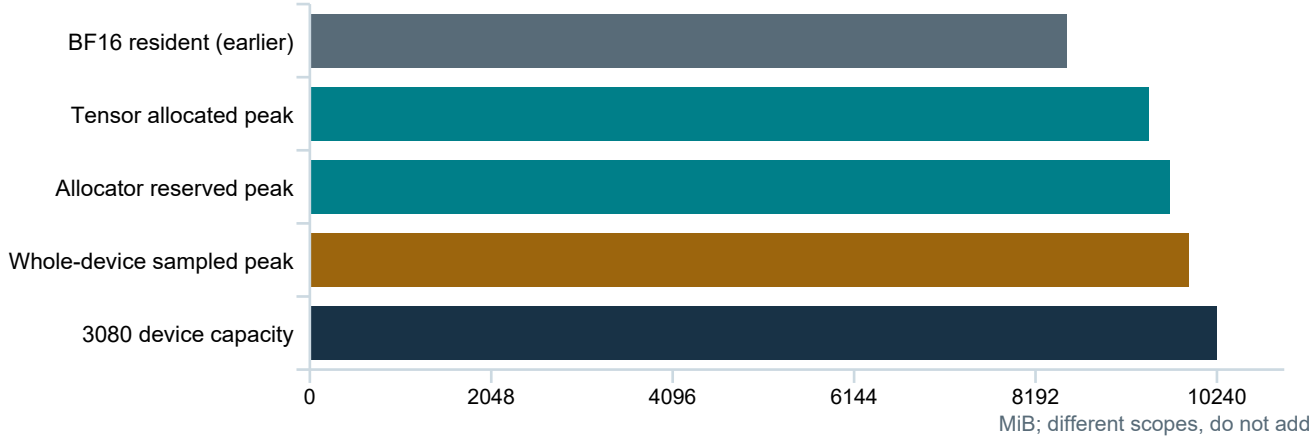
- Prior baseline/adapter A/B tests produced matching WAV hashes in eight pairs. Gains were small and variable, around 0-4% by voice; Ivan was slightly slower in its median comparison. [E2]

- No new native-speaker listening panel or identity score was performed. Selected BF16 references had prior approval; sentence overrides and all quantized variants still require listening review.

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GPU memory is the main fit problem

A component fitting alone is a different requirement from the complete application fitting.



Measurement	Value	Meaning
Earlier BF16 resident allocation	8,546 MiB	Speech model/codec tensor allocation in its measured loaded state
New voice allocated peak	9,466 MiB	Maximum recorded live CUDA tensor allocation for the voice test
New voice reserved peak	9,708 MiB	Maximum recorded caching-allocator reservation; includes allocated memory
New quick-test whole GPU peak	9,922 MiB	Sampled total GPU 0 use, including other GPU consumers
Headroom at sampled device peak	318 MiB	Only about 3.1% of the 10,240 MiB device remained

These rows are overlapping measurements, not separate memory bills. Some come from different calls or the earlier baseline, and maxima need not occur at the same instant. Whole-device sampling was approximately every 2 s, so a short higher peak may be missed. CUDA peak counters capture a different process-local scope. [E1-E3]

Why unused-cache release is limited

PyTorch distinguishes live tensor allocations from memory held by its caching allocator. Releasing unused cached blocks can return space to other applications; it does not remove resident model tensors. It can also add allocation overhead later. The setting cannot turn an approximately 8.35 GiB resident BF16 speech model into a small model. [R2]

Windows dedicated/shared process allocations are useful diagnostics, but are not identical to nvidia-smi GPU 0 totals. The kit marks missing counters as unavailable. A shared allocation on its own does not prove a particular tensor was paged or establish the cause of a slowdown.

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Transcription and smaller tuning gains

Useful refinements exist, but they are smaller than the observed model-placement penalty.

CPU multilingual transcription

Beam size	Samples	Median s	Maximum s	Selected recording
5	2	3.301	3.332	Both transcripts matched
1	2	3.042	3.052	Both transcripts matched

Beam 1 saved about 0.259 s, or 7.8%, on this one stock recording. This is a useful small result, not a broad recognition-quality score. There was no microphone, room noise, overlapping speech or accent diversity in this test. Earlier short generated clips also showed a recognition error for one Ivan phrase in both beam settings. [E1, E2]

The setup uses large-v3-turbo through faster-whisper, CPU INT8, eight CPU threads, the sr language token, Latin output and a Montenegrin prompt. sr is a supported Serbian language token used here as a proxy; it is not a separate native cnr speech-recognition model. English-only small.en is unsuitable for this requirement. [R3, R4]

Dialogue batch/thread comparisons

Configuration	Warm median s	Change versus baseline
8 threads; batch 256 / microbatch 128	3.533	Baseline
8 threads; batch 128 / microbatch 64	3.871	0.338 s slower; +9.6%
4 threads; batch 256 / microbatch 128	3.455	0.078 s faster; -2.2%

Each configuration used four requests, leaving only three warm samples. The fixed run order and small counts do not establish statistical significance. The 4-thread result is a candidate for repeated tests with the game active, not proof that four threads is always optimal. The smaller batch did not help speed in this screen.

Scale of improvement needed

Moving a 23 s reply to 14 s requires about a 39% reduction. Moving a 30 s reply to 14 s requires about 53%. A 0.08 s thread improvement or 0.26 s STT improvement cannot provide that reduction. Removing major CPU offload, excessive generations or unstable voice retries has much larger potential.

Do not add the separate dialogue, voice and STT medians to claim an integrated latency. Their workloads, answer lengths and memory conditions differ.

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The NF4 experiment: fit improved, result failed

This was a deliberate lower-precision comparison, not a replacement of the approved BF16 settings.

Setting / observation	Recorded value
Dialogue	Gemma QAT Q4; GPU-layer setting 16; 33 blocks remain on CPU
Speech	Higgs NF4; quantized cast explicitly enabled; sentence mode + adapter
Speech allocation after load	About 3,539 MiB; lower than the earlier 8,546 MiB BF16 resident figure
Whole-device sampled peak	8,455 MiB, below the 10,240 MiB device capacity
First turn	31.955 s to first prepared segment; 36.372 s for all preparation
First turn later gap	1.530 s predicted extra preparation wait
Second turn	Reply preparation failed after 88.881 s
Campaign outcome	Requested four-turn run incomplete; remaining turns not completed

Failure trace

The speech log shows one decode reaching 900 frames with stopped=false after 29.4914 s of decode, then another reaching 900 frames with stopped=false after 30.2568 s. The service returned HTTP 400 and the benchmark recorded a reply-preparation failure. The bounded audio-validation path rejected the output instead of treating it as a successful fast clip. [E1, E3]

The two capped decode phases alone consumed about 59.75 s. Other time in the failed turn includes recognition, dialogue, prefill, codec/validation and surrounding work. There is no successful second-turn first-audio timing from which to calculate a complete-run median.

Interpretation

- Lower weight memory did not remove CPU offload. The first turn still spent 24.223 s in appraisal plus dialogue, with roughly 11 generated tokens/s in the recorded LLM metrics.

- The failed stopping behavior is an observed symptom under this NF4 configuration. It is not proof of a general defect in every NF4 model or proof that the adapter caused it.

- The single 31.955 s success must not be presented as a four-turn performance result. The run failed both the <=14 s screen and completion/reliability requirements.

Recommendation: retain this option as an explicit diagnostic comparison. Do not select the tested NF4 configuration as the working low-latency preset without fixing and revalidating speech behavior.

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What the BF16 combined run tells us

Earlier functional validation deliberately moved dialogue to CPU to leave GPU space for BF16 voice.

Character	First prepared s	STT s	Voice preparation s	Audio length s
Alex	84.685	3.176	46.386	9.32
Nikola	61.062	2.989	21.279	8.36
Stefan	52.981	2.961	11.934	6.76
Ivan	56.697	2.949	15.190	8.96

All four requests completed, but the median was 58.880 s, p90 77.598 s and maximum 84.685 s. This earlier run used CPU dialogue, BF16 GPU speech, CPU STT and Rhubarb, with an earlier 24-word ordinary budget and whole-beat delivery. It is not the final 18-word/sentence preset and is not a target-4090 benchmark. [E2]

A Windows snapshot recorded the speech process at roughly 9,019 MiB dedicated plus 1,806 MiB shared GPU allocation. Alex speech preparation took about 46.386 s for a 9.32 s clip. That is much slower than the short standalone voice results and illustrates why isolated medians are not interchangeable with combined timing.

What can explain the user-reported 23-30 s replies?

Possible contributor	Evidence / confidence
CPU-offloaded dialogue blocks	Strong local evidence: partial Q8 was slow, fully offloaded QAT was much faster; exact 4090 placement still needs its logs.
Tight VRAM and competing consumers	Strong local capacity evidence; actual target-PC paging/fit has not been traced.
Longer speech and later segments	Voice time scales with generated audio/work; warm clips had little real-time margin. Longer integrated clips were slower.
Regeneration / retry	Directly observed in the NF4 failure and earlier overlength dialogue checks. Frequency on the user PC is unknown.
Only first-launch warm-up	Insufficient as a complete explanation because the user reported delays on every reply. Per-voice warm-up can still add to first uses.

The target-PC report of about 20.8 GB VRAM, 35 GB RAM and 32% CPU use was user-provided, not measured by this audit. It does not establish a memory leak or identify one bottleneck. Moderate total CPU use can still accompany a serial/thread-limited offload bottleneck; increasing thread count is not automatically a cure.

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What the optimizations actually do

Implemented behavior is separated from the evidence for a speed or quality benefit.

Change	Purpose	Evidence / practical limit
Full GPU dialogue offload where capacity permits	Avoid repeated CPU block execution	Largest observed configuration gain. Simultaneous BF16 speech changes the fit problem.
Two 4,096-token prompt slots; stable prefix; compact control JSON	Reuse appraisal/dialogue prompts and reduce repeated work	Implemented. Cache usefulness depends on prompt/history changes; not every request is a cache hit.
Flash Attention and Q8 KV cache	Reduce attention/cache cost and memory	Enabled test baseline. Q8 KV describes the cache, not dialogue weight precision.
Ordinary word target + detail exception	Reduce unnecessary tokens and speech duration	Ordinary 18-word ceiling/shorter prompt target; fuller/open requests retain larger budgets.
Explicit target below ordinary ceiling	Avoid overlength regeneration	Three earlier ordinary examples fell from 6.613-7.519 s to 3.792-4.291 s, with one dialogue generation each.
Sentence delivery	Prepare an initial part before later WAVs are ready	Can reduce first-part wait; changes pacing/prosody and may add repeated prefill.
Resident speech and reference encodings	Avoid model reload and re-encoding every turn	Helps repeated use; resident BF16 weights occupy most of a 10 GB GPU. References are cached, not a proven full reusable TTS KV prefix.
Optional decoder adapter	Reduce repeated CPU transfers / position allocations	Small variable paired gains, with matched tested WAVs. It is not a replacement speech engine.
Unused VRAM cache release	Give unused allocator blocks back to other consumers	Does not release live weights; overhead can increase.
CPU INT8 STT, optional beam 1	Reserve VRAM and reduce search work	About 3 s on the selected recording; broader recognition review required.

These changes are selectable service behavior and test settings. The 3080 kit uses scoped test configurations and does not silently rewrite the user application preset. Q8 remains the primary package model; QAT Q4 is an explicit alternative. Performance changes that alter precision, reply length or sentence prosody require quality review as well as timing. [E1-E4]

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Recorded voices, caching and language

A cloned reference recording and a complete recorded answer solve different problems.

How recorded voices help

A cloned-voice reference establishes speaker characteristics and delivery instructions; the model still has to generate new audio for new text. A bank of approved complete answers can bypass generation only when the requested speaker, text and emotion match an approved clip.

The current exact-answer bank contains five clips: three for Ivan, one for Nikola and one for Alex, with none for Stefan. It is useful for those repeated lines, not general conversation. The implementation matches complete approved lines; it does not substitute vaguely similar answers or add generic filler to conceal a slow response.

What is cached

Cache / preparation	Benefit	Limit
Voice reference encodings	Avoid repeated reference encoding	The main TTS decoder still generates every new response.
Prepared fixed opening	Fast repeat of a known opening	Does not accelerate an arbitrary follow-up answer; TTS may still be resident.
Exact approved audio bank	Can eliminate synthesis for a matching line	Only five approved lines currently; speaker/emotion/text must match.
Generated speech + mouth-cue cache	Can reuse identical prior output and compatible cues	Fresh-turn latency must be measured separately from hits; unique text will miss.
Prompt / KV reuse	Can reduce repeated LLM prompt work	Different characters, instructions and evolving history change reuse.

Fresh-generation benchmarks deliberately disable the recorded bank and speech cache. Otherwise a fast cached answer could be mistaken for a fast model. Interruption and playback acknowledgements also matter: generated words are not automatically committed as heard when audio has not played.

Montenegrin acceptance is separate from successful synthesis

Dialogue is configured for cnr with Latin text, while recognition uses the supported sr token as a proxy. A native reviewer should check regional wording, grammar, pronunciation, names and emotional suitability. Correct script and a valid WAV do not establish native-language accuracy.

Review all four assigned speakers, including fallback emotions. Sentence overrides can change an approved whole-beat performance even when BF16 precision and speaker references stay the same. NF4/INT8 need a fresh comparison of identity, emotion, stopping and intelligibility. No English-only fallback should be counted as satisfying this project. [E1-E4, R4]

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3080 10 GB, 12 GB and 4090

Capacity reasoning below is explicitly separate from measured response-time claims.

Workload	3080 10 GB	3080 12 GB	4090 24 GB
QAT Q4 dialogue alone	Measured, fast	Profile supported; unmeasured	Unmeasured here
BF16 speech alone	Measured; very tight	More capacity; unmeasured	Unmeasured here
Full QAT GPU + BF16 voice + game	Practical capacity mismatch	Practical capacity mismatch	More plausible fit; must measure full workload
Full Q8 GPU + BF16 voice + game	Unsuitable for this capacity	Unsuitable for this capacity	Tight potential fit; unverified
NF4 + partial GPU QAT	Tested profile failed	Starting profile only	Precision quality still requires approval
Verified <=14 s audible result	No	No	No

Rough planning arithmetic, not a VRAM measurement

The Q8 GGUF file is about 11.80 GiB; QAT Q4 is about 6.50 GiB. The earlier loaded BF16 speech allocation was about 8.35 GiB. Adding these proxies gives about 20.15 GiB for Q8 plus BF16, or 14.84 GiB for QAT plus BF16, before a complete dialogue/game working-memory budget.

A GGUF file size is not identical to its GPU allocation. CUDA padding, buffers, KV caches, residency choices, graphics, display use and allocator behavior must be measured. The rough Q8 sum leaves only about 3.85 GiB within a nominal 24 GiB device; it does not prove the 4090 fits or reaches 14 s. QAT leaves considerably more planning margin, with an explicit dialogue-precision tradeoff.

RAM and CPU corrections

The target CPU has been corrected to 14th generation. The local test PC is specifically a 14900K; the target SKU is not confirmed. Moving from the test machine to 64 GB RAM may help host-memory headroom and avoid system paging, but does not add dedicated GPU capacity or remove serial CPU inference work by itself.

Practical interpretation

The 3080 is useful for component testing and controlled diagnostics. If the full-size BF16 voice and dialogue quality are both mandatory with the game, the more promising routes are more GPU capacity, a second inference device/host, or an approved smaller model. The report provides no measured 3080-to-4090 speed multiplier. [E1-E3]

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Recommended next steps

Prioritize changes that address the observed bottlenecks before small system tweaks.

Priority	Action	Success check
1	On the real 4090, record model identity, actual offloaded blocks, free/dedicated/shared memory and stage timings with desktop game active.	Confirm where the current 23-30 s goes; do not infer it from total CPU %.
2	Compare full-GPU Q8 + BF16 only if it fits; use the one-slot/smaller-batch experiment deliberately.	No driver spill-related stalls, no load errors, acceptable actual first sound and later gaps.
3	If capacity is tight, compare optional full-GPU QAT Q4 + BF16 using the same language, speakers and representative prompts.	Accept only after dialogue/voice review; this is a precision alternative.
4	Warm each assigned voice/reference before timed sessions and keep workers resident when memory permits.	Record cold first-use separately; confirm warm turns still meet the target.
5	Investigate every regeneration event and overlong decode.	No capped/failed speech, no repeated invalid language/length retries.
6	Then evaluate sentence mode, beam 1, batch/thread changes and optional alignment simplification.	Measured total benefit, acceptable pacing, transcripts and lip animation.
7	If one GPU still cannot fit mandatory quality, design a split-device/host deployment.	Explicit routing, audio transport and reliability tests; not implemented by this kit.

A planning budget for the target

As an engineering objective, a 14 s audible response could allocate approximately 3 s to recognition, 4 s to appraisal/dialogue, 4 s to first speech and 3 s to endpointing, alignment, transfer and scheduling. These are **design allowances, not a measured combined prediction**. The standalone results motivate testing this budget but do not prove it is achievable concurrently.

What not to spend effort on first

The tested 2.2% thread gain, a small decoder gain or a blanket Windows power/HAGS tweak cannot explain a 39-53% required reduction. Do not overclock, disable the page file or change allocator/driver policies as an assumed cure. Use telemetry to identify a specific thermal, power, fragmentation or spill symptom before choosing a controlled system experiment.

Keep the failed NF4 preset out of normal sessions until its speech validation and stopping behavior are resolved.

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What the test download contains

The delivered ZIP is an update/test overlay, not a complete application.

Release: v0.5.4-rtx3080-test-kit. **Source:** 379d7963c0d38d1e055d7061e4617a1d84d593ff. **ZIP:** 1,328,947 bytes, with 44 manifest-listed files plus its embedded manifest. The private GitHub release has matching local/remote SHA-256 digests for ZIP, manifest and checksum. [E5]

Menu group	What it runs
Quick / tuning	Separate dialogue, voice and STT; then smaller-batch / four-thread dialogue comparisons
Voice / decoder A/B	Four BF16 voices, first/warm passes; optional matched-output adapter comparison
CPU STT / Q8	Beam 5 vs 1; primary-Q8 partial-offload comparison
Combined diagnostics	BF16 speech + CPU dialogue, or explicit experimental NF4 + partial GPU dialogue
Preflight / monitor	Hardware/runtime inventory and GPU/CPU/RAM sampling during a manual game test

How it looks when used

Close the game, run Stop Services.cmd, merge the ZIP contents into the existing application, then run Test RTX 3080.cmd and choose 1. It prints a timestamped result folder under services/.runtime/rtx3080-tests/. The console reports completed, skipped or failed for each stage.

```
REPORT.txt: readable conclusions
report.json and summary.csv: measurements and status
hardware.json: capacity and installed versions
gpu-telemetry.jsonl: resource time series
Per-stage folders: exact test configuration, logs and available WAVs
manual-game-timings.csv: blank 20-turn audible-timing sheet
```

Engineering protections

The runner does not rewrite config.local.json, download models, apply driver/power changes or upload reports. Missing assets or insufficient free-memory screening thresholds produce explicit skips. Existing service ports block isolated tests instead of being taken over. Cleanup targets owned test process trees; later failures retain partial results.

During validation, the launcher health check was corrected to recognize both llama and Higgs responses; Windows port availability uses an exclusive-bind check; CPU/RAM metrics can survive unavailable GPU counters. Regression tests cover these boundaries. Final exact-source suite: 204 passed in 15.441 s.

Prerequisites remain the complete game/runtime, assigned voice packs and required model weights. The test template is not a safe full-conversation preset simply because its dialogue-only GPU setting is 99.

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Acceptance test for a 14-second claim

A fast median is insufficient when the requirement applies to every answer.

Run order

- Capture the unchanged baseline first. Use the exact target model, speech precision, speakers, language and game settings. Record versions and keep desktop and VR results separate.

- Use isolated tests to select a candidate, then run one combined preparation test. Reject errors and invalid audio before considering speed.

- In the actual desktop game, record at least 20 new turns across all four characters: ordinary questions, a request for detail, a longer history and an interruption. Use a real therapist-question microphone input, not the stock patient-statement fixture.

- From an audio/video recording, measure the end of user speech to first audible response. Measure unintended silence between response parts separately from deliberate emotional pauses.

- Record cold startup/first voice use, then warm median, p90, maximum, success count, transcript errors and native-language/voice observations. Repeat promising settings across longer sessions; a larger repeated sample is needed to understand rare stalls.

Gate	Required evidence
Latency	Every included turn <=14 s for a claim about every measured answer; report maximum and failures, not only median.
Continuity	No unexplained preparation stalls between parts. Choose and state the allowable extra-gap threshold before comparing candidates.
Validity	No rejected/clipped/capped speech; no counting a failed turn as a fast response.
Language / identity	Native Montenegrin review and all four speaker identities/emotions acceptable.
Memory / stability	No allocation failures; sustained resource trend understood; no latency-correlated spill symptoms.
Interruption	Unplayed content stays unconfirmed; no stale audio/history after cancellation.

What remains unverified

Actual target 4090 behavior; 3080 12 GB performance; microphone endpointing and audio transport; physical speaker/headset output; target game/VR resource use; sustained long-conversation stability; and native-speaker approval of the optimized delivery. The existing player fixture check used approved recordings and validated streaming/interruption behavior, not AI response latency.

The phone connection/notifications were user-confirmed earlier, but notification delivery is not part of the performance evidence and no SMS delivery was verified.

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Evidence, confidence and final assessment

All measured findings should be traceable to the completed test campaign.

ID	Evidence / reference
E1	docs/benchmarks/2026-10-17-rtx3080-test-kit.json - new quick, voice, STT, tuning and failed NF4 results.
E2	docs/benchmarks/2026-10-17-latency-audit.json - earlier Q8/QAT comparison, paired voice checks and BF16 combined diagnostic.
E3	Local raw reports/logs in services/.runtime/: rtx3080-kit-validation-final, rtx3080-kit-tuning-validation, rtx3080-kit-nf4-final; earlier latency-audit-2026-10-17.
E4	Source 379d7963c0d38d1e055d7061e4617a1d84d593ff; exact checkout regression log: 204 tests, 15.441 s, passed.
E5	Published private v0.5.4 release; local ZIP CRC/file-hash checks; server asset SHA-256/size verification.

The 3080 quick/tuning/NF4 test records show config_local_unchanged=true and no remaining test service ports. Telemetry is sampled and sometimes has unavailable Windows GPU counters. These gaps are not filled with zero values or guessed measurements.

Primary technical references

[R1] llama.cpp b10909 model placement source - pinned output-layer/offload behavior.
[R2] PyTorch 2.8 CUDA memory management - live allocations, reserved cache and empty_cache.
[R3] faster-whisper documentation - CTranslate2/CPU INT8 usage.
[R4] Whisper language token definitions - sr language token; no separate cnr token in that list.

Confidence

High for the recorded samples: component times, the NF4 failure, memory counters within their stated scopes, configuration preservation and package hashes. **Moderate for diagnosis:** CPU offload and capacity are the clearest local contributors, but the user 4090 needs its own trace. **Unverified:** real audible 14 s, all future prompts, native language/voice quality, 12 GB behavior and extended VR stability.

Final assessment: the project now has a useful diagnostic kit and credible component-level progress. QAT dialogue is fast alone, BF16 voices work alone on the selected text, and the reports expose failures rather than hiding them. The complete quality-preserving, consistently fast spoken application remains an engineering target. On a 3080 10 GB, memory capacity is a structural obstacle; the tested NF4 workaround did not solve it.

Private GitHub release: test kit, checksums and this detailed report

```
Test ZIP SHA-256:
74be7d44cebcfd59e6f422741d05edc1f
de7f335a82c49a25d115cb6c6907b13
```